ME551 Vibrations in Linear Systems

Lecture 12

Multiple Degree of Freedom Systems 4 (Gyroscopic Syst.)

Chap 4. Multi-Degree-of-Freedom System

4.1 Equations of Motion
4.2 Geometric Representation of Motion. Equilibrium Points
4.3 Stability of Equilibrium Points. Liapunov Direct Method
4.4 Linearization about Equilibrium Points
4.5 Stability of Linearized Systems
4.6 Linear Conservative Natural Systems. Symmetric Eigenvalue Problem
4.7 Gyroscopic Conservative Systems
4.8 Nonconservative Systems. Nonsymmetric Eigenvalue Problem
4.9 Response of Discrete Systems to Harmonic Excitations
4.10 Response of Multi-DOF Systems to Arbitrary Excitations
i. Undamped natural systems
ii. Systems with proportional viscous damping
iii. Systems with small damping
iv. Systems with structural damping
v. Undamped gyroscopic systems
vi. General nonconservative systems
4.11 Discrete-Time Systems

Linearized system:

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \left(\underset{\text{sym}}{\boldsymbol{C}^*} + \underset{\text{skew sym}}{\boldsymbol{G}}\right)\dot{\boldsymbol{q}} + \left(\underset{\text{sym}}{\boldsymbol{K}} + \underset{\text{skew sym}}{\boldsymbol{H}}\right)\boldsymbol{q}(t) = 0$$

$$\downarrow$$

Gyroscopic Conservative system: $\boldsymbol{C}^* = \boldsymbol{H} = 0$

$$\downarrow$$

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{G}\dot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} = 0$$

© Prof. Yong-Hwa Park, KAIST

4.7 Gyroscopic Conservative Systems

Note:

Let $\boldsymbol{u}_r = \boldsymbol{x} + \boldsymbol{y}i$

$(\boldsymbol{x} - \boldsymbol{y}i)^T \boldsymbol{M}(\boldsymbol{x} + \boldsymbol{y}i)$
$= \boldsymbol{x}^T\boldsymbol{M}\boldsymbol{x} + \boldsymbol{x}^T\boldsymbol{M}\boldsymbol{y}i - \boldsymbol{y}^T\boldsymbol{M}\boldsymbol{x}i + \boldsymbol{y}^T\boldsymbol{M}\boldsymbol{y}$
$= \boldsymbol{x}^T\boldsymbol{M}\boldsymbol{x} + \boldsymbol{y}^T\boldsymbol{M}\boldsymbol{y} > 0$

$(\boldsymbol{x} - \boldsymbol{y}i)^T \boldsymbol{G}(\boldsymbol{x} + \boldsymbol{y}i)$: Scalar
$= \boldsymbol{x}^T\boldsymbol{G}\boldsymbol{x} + \boldsymbol{x}^T\boldsymbol{G}\boldsymbol{y}i - \boldsymbol{y}^T\boldsymbol{G}\boldsymbol{x}i + \boldsymbol{y}^T\boldsymbol{G}\boldsymbol{y}$ ...(1)
$= \left[(\boldsymbol{x} - \boldsymbol{y}i)^T \boldsymbol{G}(\boldsymbol{x} + \boldsymbol{y}i)\right]^T$: Scalar
$= (\boldsymbol{x} + \boldsymbol{y}i)^T \boldsymbol{G}^T(\boldsymbol{x} - \boldsymbol{y}i)$
$= \boldsymbol{x}^T\boldsymbol{G}^T\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{G}^T\boldsymbol{y}i + \boldsymbol{y}^T\boldsymbol{G}^T\boldsymbol{x}i + \boldsymbol{y}^T\boldsymbol{G}^T\boldsymbol{y}$ ...(2)
$\downarrow\Leftarrow (1)+(2), \boldsymbol{G} = -\boldsymbol{G}^T$
$2(\boldsymbol{x} - \boldsymbol{y}i)^T \boldsymbol{G}(\boldsymbol{x} + \boldsymbol{y}i)$
$= \boldsymbol{x}^T(\boldsymbol{G} - \boldsymbol{G}^T)\boldsymbol{y}i + \boldsymbol{y}^T(-\boldsymbol{G} + \boldsymbol{G}^T)\boldsymbol{x}i$
$\downarrow\Leftarrow \left[\boldsymbol{y}^T(-\boldsymbol{G} + \boldsymbol{G}^T)\boldsymbol{x}i\right]^T = \boldsymbol{x}^T(\boldsymbol{G} - \boldsymbol{G}^T)\boldsymbol{y}i$: Scalar
$(\boldsymbol{x} - \boldsymbol{y}i)^T \boldsymbol{G}(\boldsymbol{x} + \boldsymbol{y}i) = \boldsymbol{x}^T(\boldsymbol{G} - \boldsymbol{G}^T)\boldsymbol{y}i = 2\boldsymbol{x}^T\boldsymbol{G}\boldsymbol{y}i = gi$

Natural (homogeneous) vibration response, natural frequencies, modeshapes?

$\underset{n\times n}{\boldsymbol{M}}\,\underset{n\times 1}{\ddot{\boldsymbol{q}}} + \underset{n\times n}{\boldsymbol{G}}\,\underset{n\times 1}{\dot{\boldsymbol{q}}} + \underset{n\times n}{\boldsymbol{K}}\,\underset{n\times 1}{\boldsymbol{q}} = 0$, where $\boldsymbol{M}$: Real Sym. PD, $\boldsymbol{G}$: Real Skew sym., $\boldsymbol{K}$: Real Sym. PD

$\downarrow\Leftarrow \boldsymbol{q}(t) = \underset{n\times 1}{\boldsymbol{u}}\, e^{st}$: Assume harmonic motion: is it reasonable?

$$\left[s^2\boldsymbol{M} + s\boldsymbol{G} + \boldsymbol{K}\right]\boldsymbol{u} = 0 \Rightarrow \begin{cases} C.E.:\ \det\left[s^2\boldsymbol{M} + s\boldsymbol{G} + \boldsymbol{K}\right] = 0 \Rightarrow s_r : \text{pure imaginary?!} \\ \left[s_r^2\boldsymbol{M} + s_r\boldsymbol{G} + \boldsymbol{K}\right]\boldsymbol{u}_r = 0 \Rightarrow \boldsymbol{u}_r : \text{complex?!} \end{cases}$$

$\downarrow\Leftarrow$ premultiply by $\bar{\boldsymbol{u}}_r^T$ $\left(\equiv \boldsymbol{u}_r^H, \text{conjugate transpose of } \boldsymbol{u}_r : \text{Unknown yet!}\right)$

$$s_r^2\underbrace{\bar{\boldsymbol{u}}_r^T\boldsymbol{M}\boldsymbol{u}_r}_{Note:\ m_r>0} + s_r\underbrace{\bar{\boldsymbol{u}}_r^T\boldsymbol{G}\boldsymbol{u}_r}_{Note:\ ig_r} + \underbrace{\bar{\boldsymbol{u}}_r^T\boldsymbol{K}\boldsymbol{u}_r}_{Note:\ k_r>0} = 0 \Rightarrow m_r s_r^2 + ig_r s_r + k_r = 0 : \underbrace{s_r = \frac{-g_r \pm \sqrt{g_r^2 + 4m_r k_r}}{2m_r}\, i}_{Pure\ imaginary!}, \text{since } g_r^2 + 4m_r k_r > 0$$

Computational Problem:

$\det\left[s^2\boldsymbol{M} + s\boldsymbol{G} + \boldsymbol{K}\right] = 0 \Rightarrow$ How to get s_r for a large n?

2n solutions of root s?

© Prof. Yong-Hwa Park, KAIST

State Equation Approach for Standard EVP

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{G}\dot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} = 0:$$

$\downarrow\Leftarrow$ Augmentation by $\boldsymbol{K}\dot{\boldsymbol{q}} = \boldsymbol{K}\dot{\boldsymbol{q}}$

$$\underbrace{\begin{bmatrix} \boldsymbol{K} & 0 \\ 0 & \boldsymbol{M} \end{bmatrix}}_{\equiv \boldsymbol{M}^*:\ 2n\times 2n \text{ sym, PD}} \underbrace{\frac{d}{dt}\begin{Bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{Bmatrix}}_{\dot{\boldsymbol{x}}:\ 2n\times 1} = - \underbrace{\begin{bmatrix} 0 & -\boldsymbol{K} \\ \boldsymbol{K} & \boldsymbol{G} \end{bmatrix}}_{\equiv \boldsymbol{G}^*:\ 2n\times 2n \text{ skew sym}} \underbrace{\begin{Bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{Bmatrix}}_{\boldsymbol{x}:\ 2n\times 1}:$$

$$\boldsymbol{M}^*\dot{\boldsymbol{x}} = -\boldsymbol{G}^*\boldsymbol{x}(t)$$

$\downarrow\Leftarrow$ $\boldsymbol{x}(t) = \boldsymbol{x}e^{i\omega t}$: Assumption of harmonic motion (reasonable?)

Complex Eigenvalue Problem : of order 2n !

$$i\omega\boldsymbol{M}^*\boldsymbol{x} = -\boldsymbol{G}^*\underset{2n\times 1}{\boldsymbol{x}}$$: One symmetric and one skew-symmetric matrices

$\downarrow\Leftarrow$ $\boldsymbol{x} = \boldsymbol{y} + i\boldsymbol{z}$: Assumption of complex eigenvector ($\boldsymbol{y}$ & $\boldsymbol{z}$: real)

Real Eigenvalue Problem : of order 2n !

Re. part: $$-\omega\boldsymbol{M}^*\underset{2n\times 1}{\boldsymbol{z}} = -\boldsymbol{G}^*\underset{2n\times 1}{\boldsymbol{y}} \Rightarrow \boldsymbol{z} = \frac{1}{\omega}\boldsymbol{M}^{*-1}\boldsymbol{G}^*\boldsymbol{y}$$

Real Eigenvalue Problem : of order 2n !

Im. part: $$\omega\boldsymbol{M}^*\boldsymbol{y} = -\boldsymbol{G}^*\boldsymbol{z} \Rightarrow \boldsymbol{y} = -\frac{1}{\omega}\boldsymbol{M}^{*-1}\boldsymbol{G}^*\boldsymbol{z}$$

$\downarrow\Leftarrow$ Formulation in terms of single vector, either $\boldsymbol{y}$ or $\boldsymbol{z}$

© Prof. Yong-Hwa Park, KAIST

State Equation Approach for Standard EVP (cont.)

$$\underset{\lambda}{\omega^2}\boldsymbol{M}^*\boldsymbol{y} = \underbrace{\overset{-\boldsymbol{G}^*}{\boldsymbol{G}^{*T}}\boldsymbol{M}^{*-1}\boldsymbol{G}^*}_{\equiv \boldsymbol{K}^*}\boldsymbol{y} \quad \left(\text{or } \omega^2\boldsymbol{M}^*\boldsymbol{z} = \boldsymbol{G}^{*T}\boldsymbol{M}^{*-1}\boldsymbol{G}^*\boldsymbol{z}: \text{ Any difference between } \boldsymbol{y} \text{ and } \boldsymbol{z}?\right)$$

Real Symmetric : PD

$$\boldsymbol{M}^* \equiv \begin{bmatrix} \boldsymbol{K} & 0 \\ 0 & \boldsymbol{M} \end{bmatrix} \Rightarrow \downarrow\Leftarrow \left\{ \boldsymbol{K}^* \equiv \boldsymbol{G}^{*T}\boldsymbol{M}^{*-1}\boldsymbol{G}^* = \begin{bmatrix} 0 & \boldsymbol{K}^T \\ -\boldsymbol{K}^T & \boldsymbol{G}^T \end{bmatrix}\begin{bmatrix} \boldsymbol{K}^{-1} & 0 \\ 0 & \boldsymbol{M}^{-1} \end{bmatrix}\begin{bmatrix} 0 & -\boldsymbol{K} \\ \boldsymbol{K} & \boldsymbol{G} \end{bmatrix} = \begin{bmatrix} \boldsymbol{K}\boldsymbol{M}^{-1}\boldsymbol{K} & \boldsymbol{K}\boldsymbol{M}^{-1}\boldsymbol{G} \\ \boldsymbol{G}^T\boldsymbol{M}^{-1}\boldsymbol{K} & \boldsymbol{K}+\boldsymbol{G}^T\boldsymbol{M}^{-1}\boldsymbol{G} \end{bmatrix} = \boldsymbol{K}^{*T} \right. : \text{ Real Symmetric !}$$

$\lambda\boldsymbol{M}^*\boldsymbol{y} = \boldsymbol{K}^*\boldsymbol{y}$ $\left(\text{or } \lambda\boldsymbol{M}^*\boldsymbol{z} = \boldsymbol{K}^*\boldsymbol{z}\right)$: Eigen-problem for two $(2n \times 2n)$ real symmetric matrices

$\downarrow\Leftarrow$ Transformation $\boldsymbol{y} = \boldsymbol{Q}^{*-1}\boldsymbol{v}$, where $\boldsymbol{M}^* = \boldsymbol{Q}^{*T}\boldsymbol{Q}^*$ (Cholesky decomposition)

$\lambda\boldsymbol{v} = \underset{2n\times 2n}{\boldsymbol{A}}\ \underset{2n\times 1}{\boldsymbol{v}}$, where $\boldsymbol{A} = \boldsymbol{Q}^{*-T}\boldsymbol{K}^*\boldsymbol{Q}^{*-1}$: $2n \times 2n$ real symmetric

$\downarrow$

$\lambda_r, r = 1,2,\ldots,2n$ distinct real?(*no*!) $\left(\omega_r = \pm\sqrt{\lambda_r}\right.$: $4n$ natural frequencies? (no!) See page 13-15 for an example of $n = 2.\left.\right)$

In fact, $\lambda_{2r-1} = \lambda_{2r}$, $r = 1,2,\ldots,n$: n **repeated real** eigenvalues λ_r $\left(2n \text{ natural frequencies, } i\omega_r, -i\omega_r, r = 1,2,\ldots,n\right) \Rightarrow$ Make $\boldsymbol{v}_{2r-1}^T\boldsymbol{v}_{2r} = 0$!

$\downarrow\Leftarrow \boldsymbol{y}_r \equiv \boldsymbol{Q}^{*-1}\boldsymbol{v}_{2r-1}$ & $\boldsymbol{z}_r \equiv \boldsymbol{Q}^{*-1}\boldsymbol{v}_{2r}$ $\qquad \left(\boldsymbol{v}_{2r-1}^T\boldsymbol{v}_{2r} = 0 \Rightarrow \left(\boldsymbol{Q}^*\boldsymbol{y}_r\right)^T\boldsymbol{Q}^*\boldsymbol{z}_r = \boldsymbol{y}_r^T\boldsymbol{M}^*\boldsymbol{z}_r = 0!\right)$

$$\underset{2n\times 1}{\boldsymbol{x}_r} = \boldsymbol{y}_r + i\boldsymbol{z}_r = \boldsymbol{Q}^{*-1}\left(\boldsymbol{v}_{2r-1} + i\boldsymbol{v}_{2r}\right) \equiv \begin{Bmatrix} \underset{n\times 1}{\boldsymbol{u}_r} \\ \underset{n\times 1}{i\omega_r\boldsymbol{u}_r} \end{Bmatrix}, r = 1,2,\ldots,n \quad \Leftarrow \boldsymbol{x}(t) = \begin{Bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{Bmatrix} = \begin{Bmatrix} \boldsymbol{u}_r \\ i\omega_r\boldsymbol{u}_r \end{Bmatrix} e^{i\omega_r t}$$

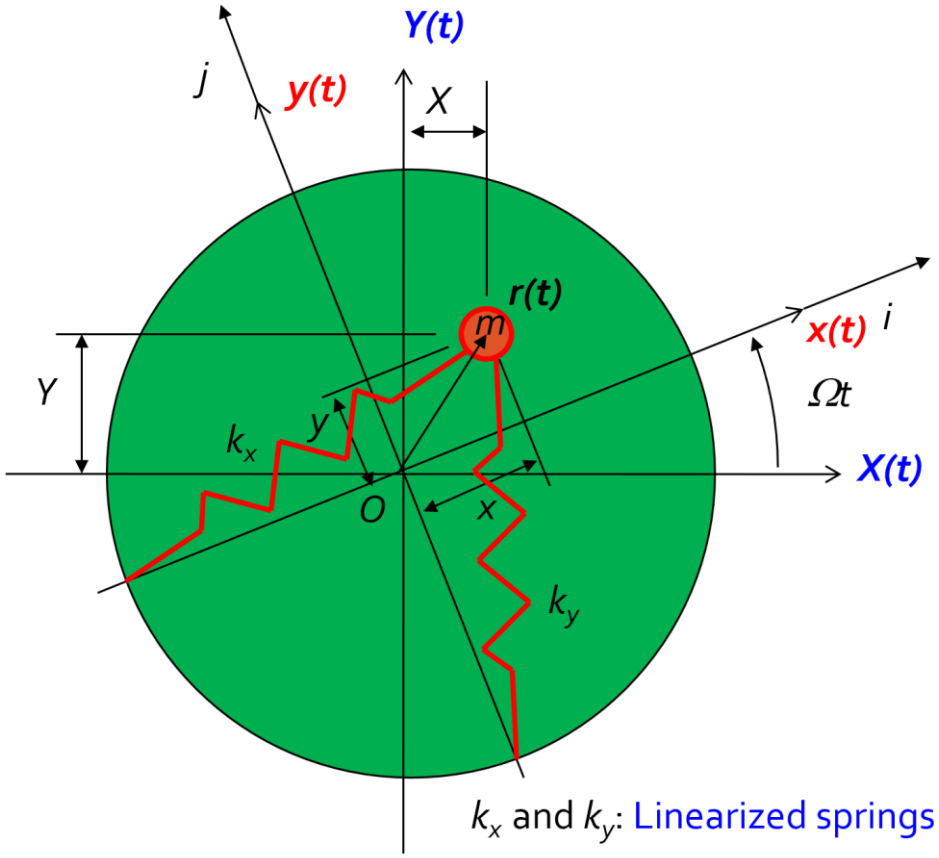
"Mode shape"

$$\therefore \begin{Bmatrix} \boldsymbol{q}(t) \\ \dot{\boldsymbol{q}}(t) \end{Bmatrix} = \boldsymbol{x}(t) = \sum_{r=1}^{n}\left(\boldsymbol{x}_r e^{i\omega_r t} + \bar{\boldsymbol{x}}_r e^{-i\omega_r t}\right) \underset{\text{Take upper half!}}{\Rightarrow} \boldsymbol{q}(t) = \sum_{r=1}^{n}\left(\boldsymbol{u}_r e^{i\omega_r t} + \bar{\boldsymbol{u}}_r e^{-i\omega_r t}\right)$$

"Mode superposition"

© Prof. Yong-Hwa Park, KAIST

Ex.4.7: Linearized Mass-Spring (Undamped) System on Rotating Disk



1) Eigenvalue, Eigenvector in closed form for $k_x=k_y=k$
2) Eigenvalue formulations for $k_x \neq k_y$
3) Solutions in Sec. 4.7 of text:
   (i) By the method specifically for a gyroscopic system

$$m = 1\ \text{kg}, \Omega = 2\,\text{rad/s}, k_x = 5\,\text{N/m}, k_y = 10\,\text{N/m}$$

$$c = h = 0$$

© Prof. Yong-Hwa Park, KAIST

1) Formulations of Eigen structure in closed form for $k_x=k_y=k$

- **Eigenvalues** and **Eigenvectors**

Gyroscopic Conservative System:

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x} \\ \ddot{y} \end{Bmatrix}+\begin{bmatrix} 0 & -2m\Omega \\ 2m\Omega & 0 \end{bmatrix}\begin{Bmatrix} \dot{x} \\ \dot{y} \end{Bmatrix}+\begin{bmatrix} k-m\Omega^2 & 0 \\ 0 & k-m\Omega^2 \end{bmatrix}\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix}=\begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\downarrow\Leftarrow \begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix}=\begin{Bmatrix} x \\ y \end{Bmatrix}e^{st} \text{ :harmonic assumption}$$

$$\begin{bmatrix} ms^2+k-m\Omega^2 & -2m\Omega s \\ 2m\Omega s & ms^2+k-m\Omega^2 \end{bmatrix}\begin{Bmatrix} x \\ y \end{Bmatrix}=\begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\downarrow\Leftarrow \omega_n^2 \equiv \frac{k}{m}, C.E.=\det[\,]=0$$

$$s^2=-(\omega_n-\Omega)^2,-(\omega_n+\Omega)^2 : s_{1,2}=\pm i(\omega_n-\Omega), s_{3,4}=\pm i(\omega_n+\Omega)$$

$$\begin{Bmatrix} x \\ y \end{Bmatrix}_{s=s_i} \propto \begin{Bmatrix} 2m\Omega s \\ ms^2+k-m\Omega^2 \end{Bmatrix}_{s=s_i} \propto \underbrace{\begin{Bmatrix} i \\ 1 \end{Bmatrix}_{s_1=i(\omega_n-\Omega)}, \begin{Bmatrix} -i \\ 1 \end{Bmatrix}_{s_2=-i(\omega_n-\Omega)}, \begin{Bmatrix} -i \\ 1 \end{Bmatrix}_{s_3=i(\omega_n+\Omega)}, \begin{Bmatrix} i \\ 1 \end{Bmatrix}_{s_4=-i(\omega_n+\Omega)}}$$

The number of modes at $\pm i(\omega_n-\Omega)$ distinctively 2 and
the mode shape vectors are complex conjugate for real physical solution! How about when $\Omega=0$?

© Prof. Yong-Hwa Park, KAIST

- **Physical Solution**?

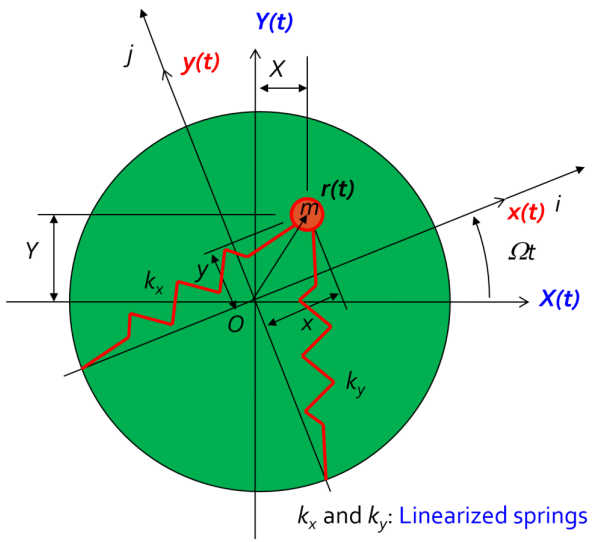
$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x} \\ \ddot{y} \end{Bmatrix} + \begin{bmatrix} 0 & -2m\Omega \\ 2m\Omega & 0 \end{bmatrix}\begin{Bmatrix} \dot{x} \\ \dot{y} \end{Bmatrix} + \begin{bmatrix} k-m\Omega^2 & 0 \\ 0 & k-m\Omega^2 \end{bmatrix}\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\downarrow\Leftarrow \begin{cases} \text{4 Eigenvectors:} \begin{Bmatrix} i \\ 1 \end{Bmatrix}_{s_1=i(\omega_n-\Omega)}, \begin{Bmatrix} -i \\ 1 \end{Bmatrix}_{s_2=-i(\omega_n-\Omega)}, \begin{Bmatrix} -i \\ 1 \end{Bmatrix}_{s_3=i(\omega_n+\Omega)}, \begin{Bmatrix} i \\ 1 \end{Bmatrix}_{s_4=-i(\omega_n+\Omega)} \\ \text{Modal summation} \end{cases}$$

$$\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} = a_1\begin{bmatrix} i \\ 1 \end{bmatrix}e^{i(\omega_n-\Omega)t} + a_1^*\begin{bmatrix} -i \\ 1 \end{bmatrix}e^{-i(\omega_n-\Omega)t} + a_2\begin{bmatrix} -i \\ 1 \end{bmatrix}e^{i(\omega_n+\Omega)t} + a_2^*\begin{bmatrix} i \\ 1 \end{bmatrix}e^{-i(\omega_n+\Omega)t}$$

$$= 2\mathrm{Re}\left[a_1\begin{bmatrix} i \\ 1 \end{bmatrix}(\cos(\omega_n-\Omega)t + i\sin(\omega_n-\Omega)t\right] + 2\mathrm{Re}\left[a_2\begin{bmatrix} -i \\ 1 \end{bmatrix}(\cos(\omega_n+\Omega)t + i\sin(\omega_n+\Omega)t\right]$$

$$= \underbrace{A_1\begin{bmatrix} -\sin[(\omega_n-\Omega)t-\phi_1] \\ \cos[(\omega_n-\Omega)t-\phi_1] \end{bmatrix}}_{\text{Forward Whirl Mode}} + \underbrace{A_2\begin{bmatrix} \sin[(\omega_n+\Omega)t-\phi_2] \\ \cos[(\omega_n+\Omega)t-\phi_2] \end{bmatrix}}_{\text{Backward Whirl Mode}}$$



A_1, ϕ_1, A_2, ϕ_2 : **Determined by 4 initial conditions** $x(0), \dot{x}(0), y(0), \dot{y}(0)$ **!**

- **Forward Whirl Mode and Backward Whirl Mode**

$$\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} = \underbrace{A_1 \begin{bmatrix} -\sin[(\omega_n - \Omega)t - \phi_1] \\ \cos[(\omega_n - \Omega)t - \phi_1] \end{bmatrix}}_{\text{Forward Whirl Mode}} + \underbrace{A_2 \begin{bmatrix} \sin[(\omega_n + \Omega)t - \phi_2] \\ \cos[(\omega_n + \Omega)t - \phi_2] \end{bmatrix}}_{\text{Backward Whirl Mode}}$$

https://www.youtube.com/watch?v=eDISfw_ts3A

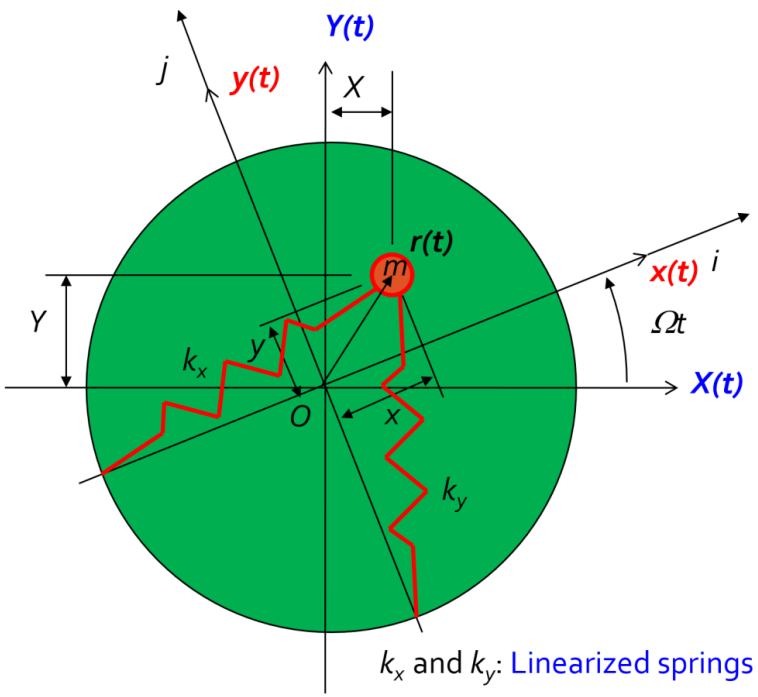
- **What happens when $\Omega = 0$?**

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x} \\ \ddot{y} \end{Bmatrix} + \begin{bmatrix} 0 & -2m\Omega \\ 2m\Omega & 0 \end{bmatrix}\begin{Bmatrix} \dot{x} \\ \dot{y} \end{Bmatrix} + \begin{bmatrix} k - m\Omega^2 & 0 \\ 0 & k - m\Omega^2 \end{bmatrix}\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\downarrow \Leftarrow \Omega = 0$$

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x} \\ \ddot{y} \end{Bmatrix} + \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

↓ Are x & y decoupled?



$$\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} = A_1 \begin{bmatrix} -\sin[(\omega_n - \Omega)t - \phi_1] \\ \cos[(\omega_n - \Omega)t - \phi_1] \end{bmatrix} + A_2 \begin{bmatrix} \sin[(\omega_n + \Omega)t - \phi_2] \\ \cos[(\omega_n + \Omega)t - \phi_2] \end{bmatrix}_{\Omega=0}$$

$$= \begin{bmatrix} -A_1 \sin(\omega_n t - \phi_1) + A_2 \sin(\omega_n t - \phi_2) \\ -A_1 \cos(\omega_n t - \phi_1) + A_2 \cos(\omega_n t - \phi_2) \end{bmatrix}$$

$$= \begin{bmatrix} A_1' \sin(\omega_n t - \phi_1') \\ A_2' \cos(\omega_n t - \phi_2') \end{bmatrix}$$

$A_1', \phi_1', A_2', \phi_2'$: **Determined by 4 initial conditions** $x(0), \dot{x}(0), y(0), \dot{y}(0)$ **!**

© Prof. Yong-Hwa Park, KAIST

- **Physical Vibration in Time Domain just at the first mode**

$$\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} = \frac{(a+ib)}{2}\begin{Bmatrix} i \\ 1 \end{Bmatrix} e^{i(\omega_n-\Omega)t} + \frac{(a-ib)}{2}\begin{Bmatrix} -i \\ 1 \end{Bmatrix} e^{-i(\omega_n-\Omega)t} + \frac{(c+id)}{2}\begin{Bmatrix} -i \\ 1 \end{Bmatrix} e^{i(\omega_n+\Omega)t} + \frac{(c-id)}{2}\begin{Bmatrix} i \\ 1 \end{Bmatrix} e^{-i(\omega_n+\Omega)t}$$

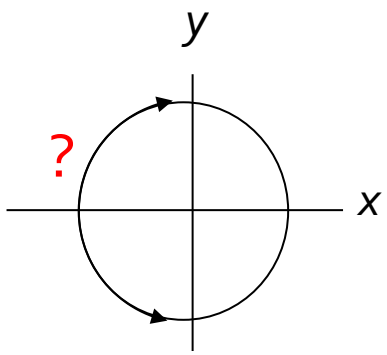
$$\downarrow\Leftarrow \begin{cases} a,b,c,d \Leftarrow \text{Initial condition:} \begin{Bmatrix} x(0) \\ y(0) \end{Bmatrix} = \begin{Bmatrix} -b+d \\ a+c \end{Bmatrix} \& \begin{Bmatrix} \dot{x}(0) \\ \dot{y}(0) \end{Bmatrix} = \begin{Bmatrix} -a(\omega_n-\Omega)+c(\omega_n+\Omega) \\ -b(\omega_n-\Omega)-d(\omega_n+\Omega) \end{Bmatrix} \\ \downarrow\Leftarrow c=d=0 \text{ for vibration at the first mode only} \left(s_{1,2} = \pm i(\omega_n-\Omega), \begin{Bmatrix} x \\ y \end{Bmatrix}_{s=s_1,s_2} = \begin{Bmatrix} i \\ 1 \end{Bmatrix}, \begin{Bmatrix} -i \\ 1 \end{Bmatrix} \right) \\ \begin{Bmatrix} x(0) \\ y(0) \end{Bmatrix} = \begin{Bmatrix} -b \\ a \end{Bmatrix} \& \begin{Bmatrix} \dot{x}(0) \\ \dot{y}(0) \end{Bmatrix} = (\omega_n-\Omega)\begin{Bmatrix} -a \\ -b \end{Bmatrix}: \begin{cases} \text{Not only speicific initial dispalcements but also} \\ \text{specific initial velocities must be provided!} \end{cases} \end{cases}$$

$$\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} = \frac{(a+ib)}{2}\begin{Bmatrix} i \\ 1 \end{Bmatrix} e^{i(\omega_n-\Omega)t} + \frac{(a-ib)}{2}\begin{Bmatrix} -i \\ 1 \end{Bmatrix} e^{-i(\omega_n-\Omega)t}$$

$$= \begin{Bmatrix} -b\cos(\omega_n-\Omega)t - a\sin(\omega_n-\Omega)t \\ a\cos(\omega_n-\Omega)t - b\sin(\omega_n-\Omega)t \end{Bmatrix}$$

$$= \begin{Bmatrix} -\sqrt{a^2+b^2}\sin\{(\omega_n-\Omega)t+\phi\} \\ \sqrt{a^2+b^2}\cos\{(\omega_n-\Omega)t+\phi\} \end{Bmatrix}, \text{ where } \begin{cases} \phi = \tan^{-1}\dfrac{b}{a} \\ \text{phase difference} = \dfrac{\pi}{2} \Rightarrow 90^\circ \text{ out of phase} \end{cases}$$

- Trajectory & direction in the *x*-*y* plane at the first mode ?



$$\left(\frac{x}{\sqrt{a^2+b^2}}\right)^2 + \left(\frac{y}{\sqrt{a^2+b^2}}\right)^2 = 1 \Rightarrow \text{ a circle:}$$

2) Eigenvalue formulations for $k_x \neq k_y$

$$\det\begin{bmatrix} ms^2 + k_x - m\Omega^2 & -2m\Omega s \\ 2m\Omega s & ms^2 + k_y - m\Omega^2 \end{bmatrix} = 0$$

$$\downarrow \Leftarrow \omega_x^2 \equiv \frac{k_x}{m},\ \omega_y^2 \equiv \frac{k_y}{m}$$

$$\det\begin{bmatrix} s^2 + \omega_x^2 - \Omega^2 & -2\Omega s \\ 2\Omega s & s^2 + \omega_y^2 - \Omega^2 \end{bmatrix} = 0$$

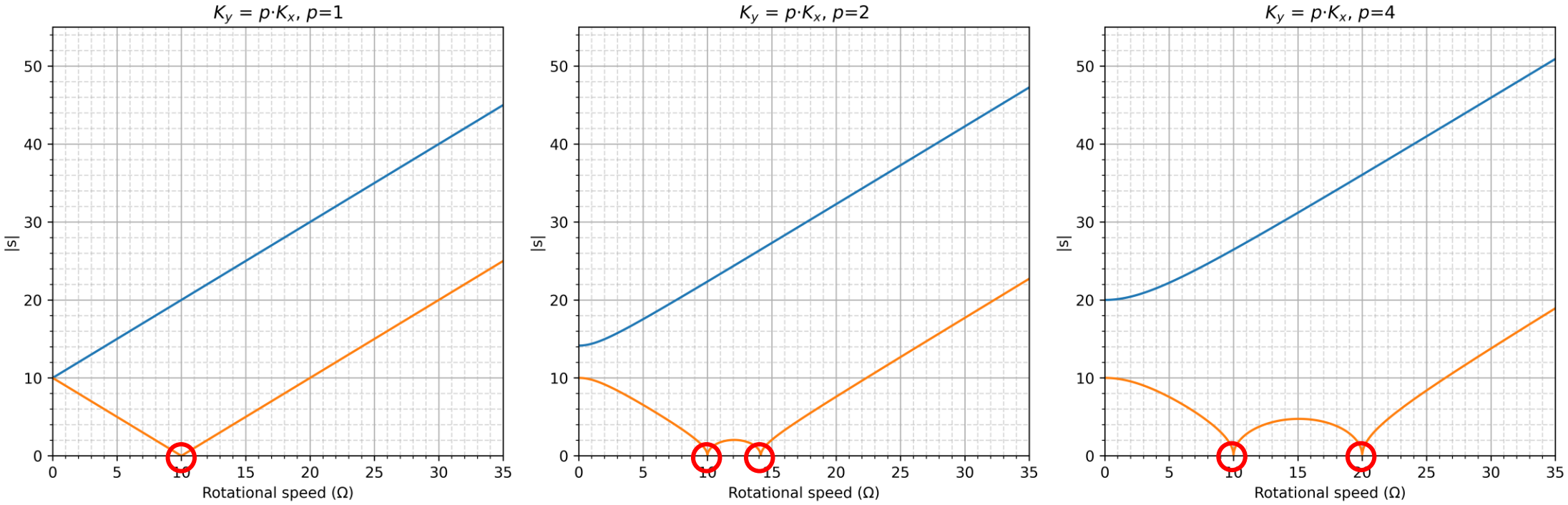
$$\downarrow \Leftarrow a \equiv \frac{1}{2}\left(\omega_x^2 + \omega_y^2\right) + \Omega^2; b \equiv \left(\omega_x^2 - \Omega^2\right)\left(\omega_y^2 - \Omega^2\right)$$

$$s_{1,2} = \pm j\sqrt{a + \sqrt{a^2 - b}}, s_{3,4} = \pm j\sqrt{a - \sqrt{a^2 - b}}$$

© Prof. Yong-Hwa Park, KAIST

Natural(Whirling) Frequency versus Rotation Speed (Campbell diagram – Wikipedia)

- Rotation-speed(Ω) dependent resonance frequencies (ω_1 & ω_2) for $p=k_y/k_x=1, 2, 4$
- Symmetric (p=1) vs. asymmetric (p=2, 4) bearing system
- Meaning of zero natural frequencies



© Prof. Yong-Hwa Park, KAIST

3) Solutions in Sec. 4.7 of text:

By the method specifically for a gyroscopic system

$$\underbrace{\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}}_{=\boldsymbol{M}} \begin{Bmatrix} \ddot{x} \\ \ddot{y} \end{Bmatrix} + \underbrace{\begin{bmatrix} 0 & -2m\Omega \\ 2m\Omega & 0 \end{bmatrix}}_{=\boldsymbol{G}} \begin{Bmatrix} \dot{x} \\ \dot{y} \end{Bmatrix} + \underbrace{\begin{bmatrix} k_x - m\Omega^2 & 0 \\ 0 & k_y - m\Omega^2 \end{bmatrix}}_{=\boldsymbol{K}} \underbrace{\begin{Bmatrix} x \\ y \end{Bmatrix}}_{=\boldsymbol{q}} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} : \boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{G}\dot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} = 0 :$$

$\downarrow \Leftarrow$ Augmentation by $\boldsymbol{K}\dot{\boldsymbol{q}} = \boldsymbol{K}\dot{\boldsymbol{q}}$

$$\underbrace{\begin{bmatrix} \boldsymbol{K} & 0 \\ 0 & \boldsymbol{M} \end{bmatrix}}_{\equiv \boldsymbol{M}^*:2n\times 2n} \underbrace{\frac{d}{dt}\begin{Bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{Bmatrix}}_{\dot{\boldsymbol{x}}} = - \underbrace{\begin{bmatrix} 0 & -\boldsymbol{K} \\ \boldsymbol{K} & \boldsymbol{G} \end{bmatrix}}_{\equiv \boldsymbol{G}^*:\ 2n\times 2n \text{ skew sym}} \underbrace{\begin{Bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{Bmatrix}}_{\boldsymbol{x}} : \boldsymbol{M}^*\dot{\boldsymbol{x}} = -\boldsymbol{G}^*\boldsymbol{x}(t)$$

$\downarrow \Leftarrow \boldsymbol{x}(t) = \boldsymbol{x}e^{i\omega t}$

$$i\omega \boldsymbol{M}^*\boldsymbol{x} = -\boldsymbol{G}^*\boldsymbol{x}$$

$\downarrow \Leftarrow \boldsymbol{x} = \boldsymbol{y} + i\boldsymbol{z}$

$$\boldsymbol{y} = -\frac{1}{\omega}\boldsymbol{M}^{*-1}\boldsymbol{G}^*\boldsymbol{z} \;\&\; \boldsymbol{z} = \frac{1}{\omega}\boldsymbol{M}^{*-1}\boldsymbol{G}^*\boldsymbol{y} : \underbrace{\omega^2}_{\lambda}\boldsymbol{M}^*\boldsymbol{y} = \underbrace{-\boldsymbol{G}^*}_{\boldsymbol{G}^{*T}}\boldsymbol{M}^{*-1}\boldsymbol{G}^*\boldsymbol{y} \text{ or } \omega^2\boldsymbol{M}^*\boldsymbol{z} = \boldsymbol{G}^{*T}\boldsymbol{M}^{*-1}\boldsymbol{G}^*\boldsymbol{z}$$

$$\downarrow \Leftarrow \boldsymbol{K}^* \equiv \boldsymbol{G}^{*T}\boldsymbol{M}^{*-1}\boldsymbol{G}^* = \begin{bmatrix} \boldsymbol{K}\boldsymbol{M}^{-1}\boldsymbol{K} & \boldsymbol{K}\boldsymbol{M}^{-1}\boldsymbol{G} \\ \boldsymbol{G}^T\boldsymbol{M}^{-1}\boldsymbol{K} & \boldsymbol{K} + \boldsymbol{G}^T\boldsymbol{M}^{-1}\boldsymbol{G} \end{bmatrix} = \boldsymbol{K}^{*T}$$

$\underbrace{\omega^2}_{=\lambda}\boldsymbol{M}^*\boldsymbol{z} = \boldsymbol{K}^*\boldsymbol{z}$: Eigen-problem for two $(2n\times 2n)$ symmetric matrices

© Prof. Yong-Hwa Park, KAIST

$$\lambda \boldsymbol{M}^* \boldsymbol{z} = \boldsymbol{K}^* \boldsymbol{z}, \text{ where } \boldsymbol{M}^* = \begin{bmatrix} \boldsymbol{K} & 0 \\ 0 & \boldsymbol{M} \end{bmatrix}, \boldsymbol{K}^* = \begin{bmatrix} \boldsymbol{K}\boldsymbol{M}^{-1}\boldsymbol{K} & \boldsymbol{K}\boldsymbol{M}^{-1}\boldsymbol{G} \\ \boldsymbol{G}^T\boldsymbol{M}^{-1}\boldsymbol{K} & \boldsymbol{K} + \boldsymbol{G}^T\boldsymbol{M}^{-1}\boldsymbol{G} \end{bmatrix}, \lambda = \omega^2$$

$$\downarrow \Leftarrow m = 1, k_x = 5, k_y = 10, \Omega = 2$$

$$\lambda \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{=\boldsymbol{M}^*} \boldsymbol{z} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & -4 \\ 0 & 36 & 24 & 0 \\ 0 & 24 & 17 & 0 \\ -4 & 0 & 0 & 22 \end{bmatrix}}_{=\boldsymbol{K}^*} \boldsymbol{z}$$

$\downarrow \Leftarrow$ Matlab(See the next page)

$$\lambda_1 = 0.2639 : \text{repeated}(\omega_1 = \pm 0.5137), \lambda_2 = 22.736 : \text{repeated}(\omega_2 = \pm 4.7682)$$

$$\boldsymbol{y}_1 = \begin{Bmatrix} -0.9835 \\ 0 \\ 0 \\ -0.1810 \end{Bmatrix}, \boldsymbol{z}_1 = \begin{Bmatrix} 0 \\ -0.3523 \\ 0.5052 \\ 0 \end{Bmatrix}, \boldsymbol{y}_2 = \begin{Bmatrix} -0.1810 \\ 0 \\ 0 \\ 0.9835 \end{Bmatrix}, \boldsymbol{z}_2 = \begin{Bmatrix} 0 \\ -0.2063 \\ -0.8630 \\ 0 \end{Bmatrix} \Rightarrow$$

$$\boldsymbol{x}_{1,2} = \boldsymbol{y}_1 \pm i\boldsymbol{z}_1 = \begin{Bmatrix} -0.9835 \\ \mp 0.3523i \\ \pm 0.5052i \\ -0.1810 \end{Bmatrix}, \boldsymbol{x}_{3,4} = \boldsymbol{y}_2 \pm i\boldsymbol{z}_2 = \begin{Bmatrix} -0.1810 \\ \mp 0.2063i \\ \mp 0.8630i \\ 0.9835 \end{Bmatrix} \Rightarrow \boldsymbol{u}_{1,2} = \begin{Bmatrix} -0.9835 \\ \mp 0.3523i \end{Bmatrix}, \boldsymbol{u}_{3,4} = \begin{Bmatrix} -0.1810 \\ \mp 0.2063i \end{Bmatrix}$$

$$\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} = a_1 \begin{Bmatrix} -0.9835 \\ -0.3523i \end{Bmatrix} e^{i0.5137t} + \bar{a}_1 \begin{Bmatrix} -0.9835 \\ +0.3523i \end{Bmatrix} e^{-i0.5137t} + a_2 \begin{Bmatrix} -0.1810 \\ -0.2063i \end{Bmatrix} e^{i4.7682t} + \bar{a}_2 \begin{Bmatrix} -0.1810 \\ +0.2063i \end{Bmatrix} e^{-i4.7682t}$$

© Prof. Yong-Hwa Park, KAIST

Calculations for two symmetric matrices by matlab

```
>> KS=[1 0 0 -4;0 36 24 0;0 24 17 0;-4 0 0 22]
>> MS=[1 0 0 0;0 6 0 0;0 0 1 0;0 0 0 1]
>> [YZ,D]=eig(KS,MS)
YZ =
   -0.9835         0   -0.1810         0
         0   -0.3523         0   -0.2063
         0    0.5052         0   -0.8630
   -0.1810         0    0.9835         0
D =
    0.2639         0         0         0
         0    0.2639         0         0
         0         0   22.7361         0
         0         0         0   22.7361
>> YZ'*MS*YZ
ans =
    1.0000         0         0         0
         0    1.0000         0    0.0000
         0         0    1.0000         0
         0    0.0000         0    1.0000
```

$$\lambda \boldsymbol{M}^* \boldsymbol{z} = \boldsymbol{K}^* \boldsymbol{z}, \text{ where } \boldsymbol{M}^* = \begin{bmatrix} \boldsymbol{K} & 0 \\ 0 & \boldsymbol{M} \end{bmatrix}, \boldsymbol{K}^* = \begin{bmatrix} \boldsymbol{K}\boldsymbol{M}^{-1}\boldsymbol{K} & \boldsymbol{K}\boldsymbol{M}^{-1}\boldsymbol{G} \\ \boldsymbol{G}^T\boldsymbol{M}^{-1}\boldsymbol{K} & \boldsymbol{K} + \boldsymbol{G}^T\boldsymbol{M}^{-1}\boldsymbol{G} \end{bmatrix}, \lambda = \omega^2$$

$$\downarrow \Leftarrow m = 1, k_x = 5, k_y = 10, \Omega = 2$$

$$\Leftarrow \lambda \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{=\boldsymbol{M}^*} \boldsymbol{z} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & -4 \\ 0 & 36 & 24 & 0 \\ 0 & 24 & 17 & 0 \\ -4 & 0 & 0 & 22 \end{bmatrix}}_{=\boldsymbol{K}^*} \boldsymbol{z}$$

$$\longrightarrow \omega_i = \pm\sqrt{\lambda_i}$$

© Prof. Yong-Hwa Park, KAIST

Highlight 12

- The gyroscopic effect in the rotating systems leads us to have skew-symmetric term in the equation of motion.

- The gyroscopic system has complex modes: first slower natural frequency of forward whirl mode and second faster natural frequency of backward whirl mode.

- The natural frequency of forward and backward modes are changing w.r.t. rotating speed of the system.

- The eigenvalue (natural frequency) can be found with standard eigenvalue problem solver by state-space representation of the gyroscopic systems.

© Prof. Yong-Hwa Park, KAIST